

Project 2: Agent-Based Simulation of *Ophiocordyceps unilateralis* infection in a Carpenter Ant Colony

Damiane Kapanadze, Hugh Richards, Andre Llaneta

1 Introduction

Ophiocordyceps unilateralis is a parasitic fungus that targets carpenter ants. When a foraging ant picks up fungal spores from the forest floor, the fungus hijacks the ant's nervous system over the course of several weeks. The ant eventually climbs to a fixed height above the ground, bites into a leaf vein in a rigid "death grip," and dies. The fungus then erupts from the ant's head and rains new spores downward onto the foraging trails below, ready to infect the next wave of ants. [1]

This system is a positive feedback loop: more infected ants eventually mean more sporulating corpses, and more sporulating corpses mean more spores. Whether this loop runs away and kills the colony, or burns itself out, depends on a race between how fast the infection spreads and how fast the colony reproduces.

Rather than writing a single equation for the average number of infected ants, as in a classical epidemic model, we track every individual ant separately. Each ant is a software object with its own state (egg, larva, adult, infected, zombie, sporulating, or dead) and its own age. At each simulated day, every ant makes its own probabilistic decisions: did this forager pick up a spore today? Did this infected ant succumb overnight? This style of simulation is called agent-based because each ant acts independently, and the colony-level outcome emerges from the combined behavior of all of them. It is also stochastic because the outcomes are random: two runs of the simulation with identical starting conditions but different random seeds can produce different histories.

Our goal is to understand what determines whether a colony survives a cordyceps outbreak. We examine the effect of three quantities: the infection rate p (how easily spores infect a forager), the initial colony size N_0 , and immune resistance (how long an infected ant can fight off the parasite before succumbing). We also compare the stochastic agent-based model against a deterministic ODE counterpart. Figure 1 shows a normal colony before any infection is introduced.

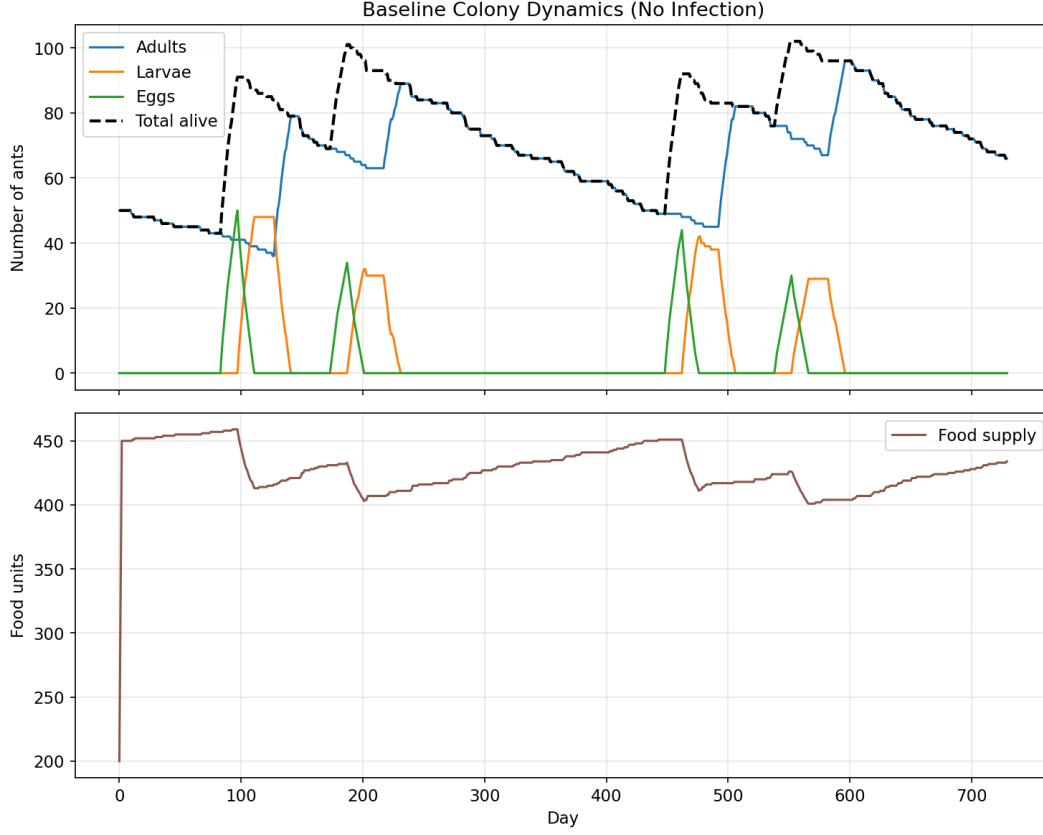


Figure 1: Normal colony dynamics with no infection ($N_0 = 50$ adults, $F_0 = 200$ food units, run for 730 days).

Adults decline gradually between the yearly egg-laying pulses. Larvae and eggs appear at those pulses and then mature according to their fixed developmental schedules, while food in the lower panel remains comfortably above zero throughout.

2 Governing Algorithm

This simulation has no differential equations in its overall agent-based algorithm because every event in the model occurs on whole-day time steps. Time advances in fixed one-day increments, and each daily phase is applied in the same order. In other words, the model is mainly controlled by the daily update rules, not by continuous equations of motion.

2.1 The unknowns

At the start of day n , the quantities being tracked are:

- The state and age of every ant in the simulation. The state s_i of ant i is one of

$$s_i \in \{\text{egg}, \text{larva}, \text{adult}, \text{infected}, \text{zombie}, \text{sporulating}, \text{dead}\}.$$

Each ant also carries d_i , the number of days it has spent in its current state.

- The **food supply** $F_n \geq 0$, measured in food units. One food unit feeds one ant for one day.
- The **environmental spore concentration** $c_n \geq 0$, a nondimensional number representing the density of viable spores in the foraging area. It rises when sporulating corpses are present and decays otherwise.

Figure 2 summarizes all possible ant states and the transitions between them.

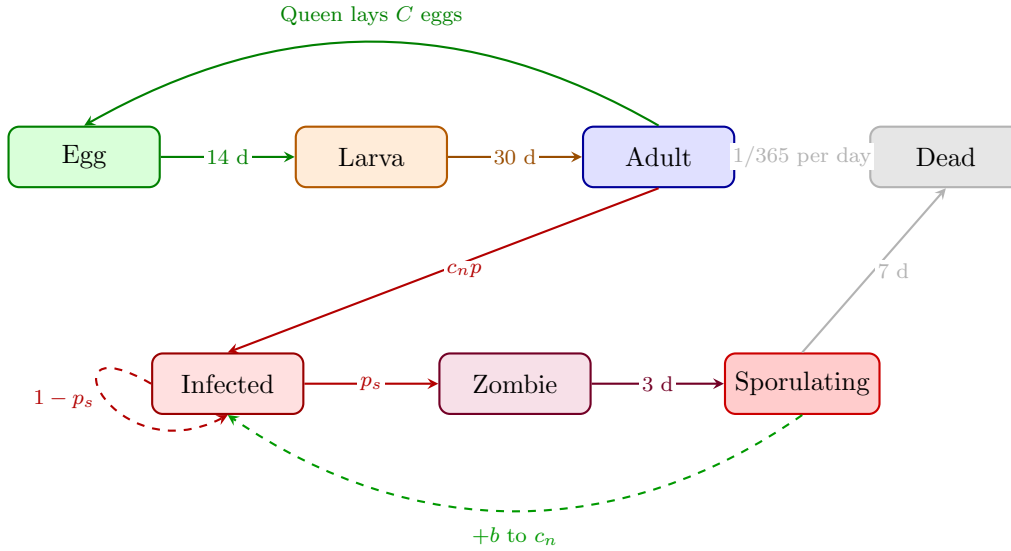


Figure 2: Individual ant state diagram.

The top row shows the normal lifecycle of an ant, from egg to adult and eventually to natural death. The bottom row shows the infection pathway. Infected ants continue to forage and eat, so they remain indistinguishable from healthy ants within the colony. The dashed green arrow represents the positive feedback loop where sporulating corpses increase c_n , which in turn drives new infections.

2.2 Fixed parameters

The following quantities are inputs to the model. They are not computed by the simulation but are chosen before it begins. We separate parameters that represent biological timing assumptions from parameters that are chosen model settings. This distinction is important because the goal is not to predict exact numbers for a real forest colony, but to build a simplified model whose assumptions are clear and whose behavior can be tested by changing the parameters.

Symbol	Value	Meaning	Rationale
δ	1/365	Daily natural-death probability for any non-egg ant	Means that, on average, a non-egg ant that avoids infection and starvation lives about one year. This is a modeling assumption used to keep the no-infection colony approximately stable, not a measured field value.
T_{egg}	14 days	Incubation time from egg to larva	Order-of-magnitude consistent with reported carpenter-ant egg-development times.
T_{larva}	30 days	Effective development time from larva to adult	Chosen so that egg-to-adult development falls in the broad 6–12 week range reported for carpenter ants when the omitted pupal stage is folded into the larval period.
T_{zombie}	3 days	Time as zombie before sporulating	Represents the short manipulated-behavior stage before the ant becomes a spore source.
T_{spore}	7 days	Duration of sporulation before the corpse is spent	Models a finite spore-release window rather than a permanently infectious corpse.

Table 1: Biological and structural timescale parameters.

Symbol	Value	Role	Interpretation / test
a	15	Foraging constant	Calibrated so that the baseline colony can usually maintain its food supply without unbounded growth.
k_{egg}	8	Egg-laying scaling constant	Calibrated together with the food model so that reproduction responds to food availability while preserving seasonal pulses.
b	0.005	Spores added per sporulating ant per day	Controls the strength of the positive spore feedback loop. Its effect is tested through infection outcomes.
k	0.30	Daily fractional decay of spore concentration	Gives a spore half-life of about 1.94 days, so spores clear quickly unless sporulating ants replenish them.
p	varied	Infection probability per unit spore concentration	Main susceptibility parameter; swept in Section 6.
p_s	0.05	Daily probability that an infected ant succumbs	Gives an expected infected period of $1/p_s = 20$ days; varied in the immune-resistance sweep.
c_0	0.05	Initial spore concentration on the introduction day	Represents a small initial contamination event. At $p = 0.15$, the initial daily infection probability is $c_0 p = 0.0075$ per healthy adult.

Table 2: Chosen model parameters. These are not exact biological measurements as they are settings used to explore how the simulated colony behaves under different infection conditions.

The biological motivation for the lifecycle and infection pathway comes from two kinds of sources. Carpenter-ant development and seasonal reproduction justify using eggs and larvae that take time to mature, along with two egg-laying windows per year [4, 5]. The death-grip, manipulated behavior, and post-mortem fungal growth motivate the infected–zombie–sporulating pathway [3, 1]. However, the model is not meant to make exact numerical predictions for a real forest colony. Parameters such as a , b , k , k_{egg} , p , and p_s are chosen model settings. We use them to create clear cases such as recovery, partial decline, and collapse, and then test how the results change when key parameters are varied. Therefore, our conclusions describe the behavior of this simulation model rather than exact outcomes in nature.

2.3 Parameter checks and derived interpretations

Several parameters in Tables 1 and 2 are easier to interpret through the quantities they imply.

First, the daily natural-death probability is

$$\delta = \frac{1}{365}.$$

The expected waiting time for natural death is therefore

$$E[T_{\text{natural death}}] = \frac{1}{\delta} = 365 \text{ days}.$$

So, in the model, natural death acts on about a one-year time scale.

Second, the infected-to-zombie transition is modeled as a repeated daily chance event. Each infected ant has probability p_s of becoming a zombie each day, so the expected waiting time is $1/p_s$. So if an infected ant succumbs each day with probability p_s , then

$$E[T_{\text{succumb}}] = \frac{1}{p_s}.$$

For the baseline value $p_s = 0.05$,

$$E[T_{\text{succumb}}] = \frac{1}{0.05} = 20 \text{ days}.$$

To test immune resistance, we run the model with different values of p_s . A smaller p_s means infected ants resist the fungus longer. For example, $p_s = 0.01$ gives an average infected period of 100 days, while $p_s = 0.15$ gives an average infected period of about 6.67 days.

Third, the spore concentration decays when no sporulating ants are adding new spores. The full update rule is

$$c_{n+1} = c_n(1 - k) + bS_n,$$

where S_n is the number of sporulating ants. If there are no sporulating ants, then $S_n = 0$, so the new-spore term bS_n disappears:

$$c_{n+1} = c_n(1 - k).$$

This means the concentration is multiplied by the same factor each day. Therefore, after n days,

$$c_n = c_0(1 - k)^n.$$

With $k = 0.30$, the half-life solves

$$(0.70)^n = \frac{1}{2}, \quad n = \frac{\ln(1/2)}{\ln(0.70)} \approx 1.94 \text{ days.}$$

Thus, environmental spores disappear quickly unless new sporulating ants keep adding spores.

Fourth, the initial contamination level $c_0 = 0.05$ creates only a small chance of infection for each ant. At the baseline infection rate $p = 0.15$, the initial daily infection probability for one healthy adult is

$$P(\text{infection}) = c_0 p = (0.05)(0.15) = 0.0075.$$

So each adult has less than a one-percent chance of infection on the introduction day. If there are about 100 healthy adults near day 200, the expected number of new infections on that day is approximately

$$100(0.0075) = 0.75.$$

This explains why the outbreak can fizzle in some stochastic trials as the initial contamination event is intentionally weak.

Finally, the spore birth value $b = 0.005$ can be interpreted by asking how much spore concentration one sporulating ant can build up during its seven-day sporulation period. Suppose one ant is sporulating for seven days and no other ants are adding spores. Each day, that ant adds b to the spore concentration, while the existing spores are multiplied by $(1 - k)$ because of decay. Starting from zero spores, the concentration after seven days is approximately

$$b(1 + (1 - k) + (1 - k)^2 + \cdots + (1 - k)^6).$$

This is a geometric sum:

$$b \sum_{j=0}^6 (1 - k)^j = b \frac{1 - (1 - k)^7}{k}.$$

With $b = 0.005$ and $k = 0.30$,

$$0.005 \frac{1 - 0.70^7}{0.30} \approx 0.0139.$$

At the baseline infection rate $p = 0.15$, this concentration would give a daily infection probability of about

$$(0.0139)(0.15) \approx 0.0021,$$

or about 0.21% for each healthy adult. This shows that a single sporulating corpse creates only a weak infection pressure by itself. The outbreak becomes dangerous when many infected ants become sporulating around the same time, because their spore contributions add together.

2.4 The daily update: seven steps

On each day, the following steps are applied in order to advance from day n to day $n + 1$.

Step 1: Lifecycle transitions. Any egg that has been an egg for at least $T_{\text{egg}} = 14$ consecutive days becomes a larva. Any larva that has been a larva for at least $T_{\text{larva}} = 30$ consecutive days becomes an adult.

Step 2: Foraging. Each adult ant and each infected adult goes out and brings back food. The amount brought back by forager i is a random draw from an exponential distribution:

$$F_i \sim \text{Exp}(\text{mean} = aN_n^{-1/3}).$$

The mean per-forager yield is modeled as $aN_n^{-1/3}$, so each individual forager brings back less food as the colony becomes larger. This represents diminishing returns. As more ants forage in the same environment, nearby food becomes harder to find or is shared among more foragers. The exponent $-1/3$ is a simplifying modeling choice rather than a measured biological constant. All food is pooled and then capped:

$$F \leftarrow \min\left(F_n + \sum_{\text{foragers}} F_i, F_{\max}\right).$$

Step 3: Eating. Every larva, adult, and infected adult consumes one food unit. Eggs do not eat, while zombies and sporulating ants have already stopped participating in colony life. If there is not enough food for everyone, a randomly chosen subset of the hungry ants is fed and the rest die of starvation immediately.

Step 4: Natural death. Every larva, adult, and infected adult dies from natural causes with probability $\delta = 1/365$, independently. This represents ordinary deaths unrelated to infection or starvation, such as aging, predation, or accidents. The value $\delta = 1/365$ means that such an adult lives about one year on average in the model.

Step 5: Egg-laying. The queen lays eggs twice a year: in spring (days 83–96 of the annual cycle) and in summer (days 173–186). The number of eggs laid each day within a window is

$$C = \left\lfloor \min\left(k_{\text{egg}} \frac{F_n}{N_n} \div 14, \frac{100}{14}\right) \right\rfloor.$$

Dividing by 14 spreads the seasonal clutch (i.e., the group of eggs laid during one seasonal egg-laying period) evenly across the window. The F_n/N_n ratio ties reproduction to colony health: a colony short on food or overburdened with mouths lays fewer eggs. The cap of 100 total eggs per clutch prevents runaway reproduction.

Step 6: Infection and spore dynamics. This step is only active after spores have been introduced. It has four sub-steps.

(a) *New infections.* Each healthy adult that is currently foraging is exposed to the spore-laden environment. It becomes infected with probability

$$P(\text{infected}) = \min(c_n p, 1).$$

The probability is proportional to both the ambient spore level c_n and the susceptibility parameter p . The minimum with 1 prevents the expression from exceeding a valid probability at high concentrations.

(b) *Immune mini-game.* Each infected ant independently “rolls” each day. With probability p_s it succumbs and transitions to zombie; with probability $1 - p_s$ it remains infected for another day. There is no recovery. Biologically, this models the immune system slowing but not stopping fungal growth.

(c) *Zombie and sporulation timers.* Any zombie that has been a zombie for $T_{\text{zombie}} = 3$ days transitions to sporulating. Any sporulating ant that has been sporulating for $T_{\text{spore}} = 7$ days transitions to dead.

(d) *Spore concentration update.* Let S_n be the number of currently sporulating ants. The concentration for the next day is

$$c_{n+1} = c_n(1 - k) + bS_n.$$

The first term models spore decay by a factor $k = 0.30$ per day. The second term adds $b = 0.005$ units of fresh spore per sporulating corpse.

Step 7: Age advance. Every ant increments its d_i value, the number of days in its current state, by one. This is what triggers lifecycle transitions in Step 1 on subsequent days.

The key behavioral differences between the infection-related states are:

State	Forages?	Eats?	Adds spores to environment?
Adult	Yes	Yes	No
Infected	Yes	Yes	No
Zombie	No	No	No
Sporulating	No	No	Yes
Dead	No	No	No

An infected ant behaves exactly like a healthy adult: it forages, eats, and spreads no spores itself. The colony has no way to detect or isolate it. Only when the ant succumbs and becomes a zombie does it withdraw from colony life.

3 Numerical Method

As the governing algorithm already advances time in discrete steps of exactly one day, there are no differential equations that need to be discretized for the agent-based simulation. Here, the one-day timestep is not an approximation introduced after writing continuous equations; it is built into the model by assumption. The algorithm is the numerical method.

The one exception is the spore concentration update

$$c_{n+1} = c_n + bS_n - kc_n,$$

which can be interpreted as forward Euler applied to the continuous ODE

$$\frac{dc}{dt} = bS(t) - kc(t)$$

with $\Delta t = 1$ day. Since the spore half-life is about 1.94 days for $k = 0.30$, a one-day timestep resolves the decay process reasonably well for the purposes of this simplified model.

We also choose the order in which the daily phases are applied. We apply foraging before eating, meaning a forager's haul contributes to the food pool on the same day the colony feeds from it. This is consistent with an interpretation where ants leave at dawn and return before feeding time.

3.1 Deterministic ODE model

To compare the stochastic simulation with a deterministic model, we replace random individual events with their average effects. In the agent-based model, each healthy adult is checked separately for infection. In the ODE model, we instead use the expected number of new infections per day, cpA , because each of the A healthy adults has infection probability cp . Similarly, instead of giving each egg its own fixed 14-day timer, the ODE model moves eggs into the larva stage continuously at rate E/T_{egg} .

This deterministic model is meant to approximate the average behavior of the agent-based simulation. It is not an exact copy because it does not track individual ants or random events. The stochastic model has integer ants, fixed state timers, random food draws, food-cap interactions, starvation events, and hard extinction. The ODE has fractional ants, exponential-style stage transitions, and smooth state variables. Therefore, we expect the two models to agree best for large populations before extinction and to disagree for small populations or after a crash.

The resulting ODEs track eight state variables:

$$\frac{dE}{dt} = \text{laying_rate}(t) - \frac{E}{T_{\text{egg}}}, \quad (1)$$

$$\frac{dL}{dt} = \frac{E}{T_{\text{egg}}} - \frac{L}{T_{\text{larva}}}, \quad (2)$$

$$\frac{dA}{dt} = \frac{L}{T_{\text{larva}}} - \frac{A}{365} - cpA, \quad (3)$$

$$\frac{dI}{dt} = cpA - p_s I - \frac{I}{365}, \quad (4)$$

$$\frac{dZ}{dt} = p_s I - \frac{Z}{T_{\text{zombie}}}, \quad (5)$$

$$\frac{dP}{dt} = \frac{Z}{T_{\text{zombie}}} - \frac{P}{T_{\text{spore}}}, \quad (6)$$

$$\frac{dc}{dt} = bP - kc, \quad (7)$$

$$\frac{dF}{dt} = (A + I)aN^{-1/3} - (L + A + I), \quad F \leq F_{\text{max}}. \quad (8)$$

Here

$$N = E + L + A + I$$

is the living colony size in the deterministic model. The laying rate uses the same seasonal windows as the stochastic simulation.

Equations (1)–(3) describe the healthy colony, where eggs are laid seasonally, eggs hatch after an average time T_{egg} , larvae mature after an average time T_{larva} , and adults die naturally at rate $1/365$ per day while also being lost to infection at rate cp .

Equations (4)–(6) describe the infection pathway. Infected ants transition to zombies at rate p_s , zombies become sporulating corpses after an average time T_{zombie} , and sporulating corpses stop producing spores after an average time T_{spore} .

Equation (7) governs spore concentration. Sporulating corpses emit spores at rate b , while spores decay at rate k .

Equation (8) tracks the food stockpile. Adults and infected ants forage at an expected rate proportional to $aN^{-1/3}$, while larvae, adults, and infected ants consume food. After each Euler step, F is projected back into the interval $[0, F_{\text{max}}]$ to match the food cap in the stochastic model.

We advance the ODE system with forward Euler at $\Delta t = 1$ day, matching the timestep of the stochastic model. The key difference is that the ODE allows fractional ants, while the stochastic model enforces integer counts and absorbing extinction. Because of these differences, the ODE is useful for checking whether the stochastic model follows a reasonable average trend early in the simulation. However, it cannot replace the agent-based model because it does not capture individual random events or permanent extinction.

4 Validation

4.1 Baseline colony with no infection

The fundamental check is to run the simulation with infection disabled and verify that colony dynamics match the intended biological and modeling expectations. Figure 1 shows a 730-day no-infection run. The results are qualitatively correct:

- Adults decline slowly between seasons at a rate consistent with $\delta = 1/365$ per day.
- Two egg-laying pulses appear per year, near days 90 and 180.
- Larva peaks follow egg peaks after the 14-day egg stage.
- Adult increases follow larva peaks after the 30-day larval stage.
- Food never drops to zero, confirming that the foraging model supports a colony of this size.

- The colony size after two years is approximately the same scale as the starting size, consistent with a roughly stable baseline population.

4.2 Food balance at every step

On any day in which no ant starves, the food update is fully determined once the random foraging amounts are drawn:

$$F_{n+1} = \min(F_n + \Sigma_{\text{foraged}} - N_{\text{eating}}, F_{\text{max}}).$$

We verified this identity at every time step of several no-infection runs by recording the food before and after the step and comparing against the sum of drawn foraging values and the number of eating ants.

4.3 Stochastic variability across runs

Figure 3 shows 50 independent runs of the same scenario ($p = 0.15$, $c_0 = 0.05$, spores introduced on day 200, $N_0 = 50$). The spread illustrates the fundamental randomness of the model. Some realizations see the outbreak fizzle early, while others sustain elevated infection for months. This is not an error and it is the expected behavior of a stochastic model. However, it also means that a single run is not representative of the model's average behavior.

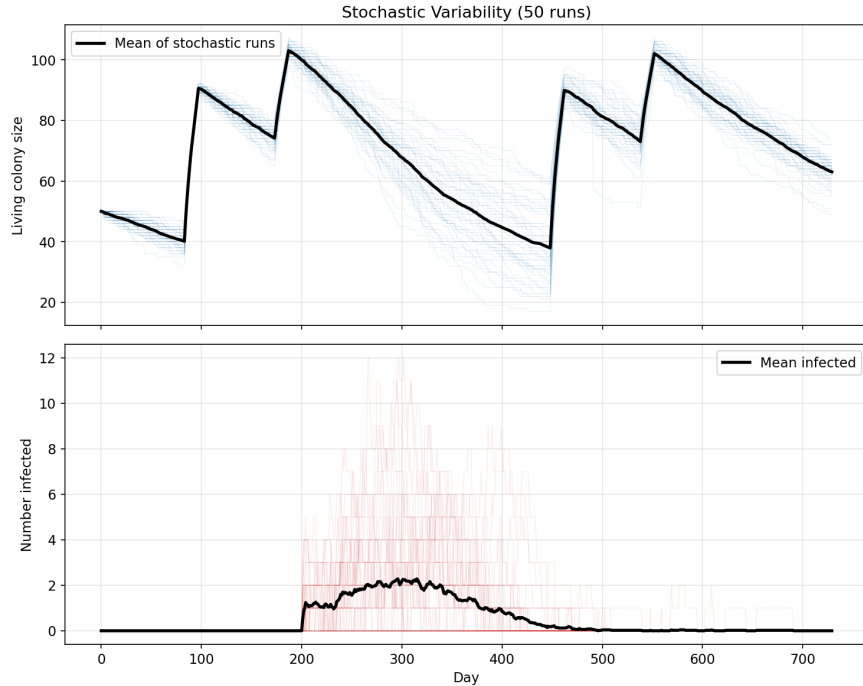


Figure 3: Living colony size across 50 independent runs at $p = 0.15$, $c_0 = 0.05$, with spores introduced on day 200. The corresponding ensemble means and standard deviations are reported in Table 3.

To avoid judging the stochastic model from only one random run, we summarize all 50 runs together. Let $X_j(n)$ be the living colony size on day n in trial j . For example, $X_3(300)$ is the living colony size on day 300 in the third simulation. If there are $M = 50$ trials, then the average living colony size on day n is

$$\bar{X}(n) = \frac{1}{M} \sum_{j=1}^M X_j(n).$$

We also compute the sample variance

$$s_X^2(n) = \frac{1}{M-1} \sum_{j=1}^M (X_j(n) - \bar{X}(n))^2,$$

which measures how much the 50 runs differ from one another on that day. The same calculation can be done for infected ants, sporulating ants, food, and spore concentration.

The resulting means and standard deviations are summarized in Table 3, so the stochastic behavior is supported numerically rather than only by visual inspection. These values were computed directly from the daily output arrays of the 50 simulations. For each selected day, the living and infected counts from all 50 trials were averaged, and the sample standard deviation was computed using $M - 1$ in the denominator. The table uses the raw simulation outputs from those days and the values were not averaged across nearby days or otherwise smoothed.

Day	Living mean	Living SD	Infected mean	Infected SD
200	99.94	2.89	0.00	0.00
250	84.34	5.04	1.56	1.75
300	67.74	8.32	2.18	2.66
365	51.24	11.62	1.32	1.80
500	81.24	6.12	0.04	0.20
730	63.02	4.87	0.00	0.00

Table 3: Selected ensemble statistics from 50 stochastic trials with $p = 0.15$, $c_0 = 0.05$, spores introduced on day 200, and $N_0 = 50$. Standard deviations are reported instead of variances so that the spread is measured in ants rather than ants squared. The final survival probability was 1.00 and the final extinction probability was 0.00.

Although the initial adult population was $N_0 = 50$, the living population at day 200 is larger because the colony has already passed through the spring and summer egg-laying windows before spores are introduced. At the spore introduction time, the mean living colony size was 99.94 ants, with almost no infected ants recorded. The mean infected count peaked near day 300 at 2.18 ants, while the largest spread in living colony size occurred near day 365, where the mean was 51.24 ants and the standard deviation was 11.62 ants. This shows that the outbreak does not affect every run equally: some colonies decline more sharply than others during the recovery-or-decline period after infection has had time to spread. By the final recorded day, the mean living colony size was 63.02 ants, the infected mean had fallen to 0, and all 50 trials still had living ants.

These validation checks address the main sources of uncertainty in the model. The stochastic simulations are not interpreted from a single trajectory; they are summarized using ensemble means and standard deviations. The deterministic ODE is not treated as an exact copy of the agent-based simulation, but as a smooth average version that should match best when the colony is large and

before extinction occurs. The parameters are also separated into biological timing assumptions and chosen model settings, with calculations such as expected waiting times, spore half-lives, and initial infection probabilities used to explain what the values mean.

5 Interesting Aspects of the Code

5.1 Behavior rules on the Ant object

Each ant's behavior is determined by three Boolean methods on the `Ant` class, encoding the rules from the table in Section 2:

```
1 def is_alive(self):
2     # False for dead, sporulating, and zombie.
3     # Sporulating ants still release spores, so they are
4     # handled separately.
5     return self.state not in (AntState.DEAD,
6                               AntState.SPORULATING,
7                               AntState.ZOMBIE)
8
9 def consumes_food(self):
10    # Larvae, healthy adults, and infected adults eat.
11    # Zombies have stopped caring about food.
12    return self.state in (AntState.LARVA,
13                          AntState.ADULT,
14                          AntState.INFECTED)
15
16 def can_forage(self):
17    # An infected ant still goes out to forage.
18    return self.state in (AntState.ADULT, AntState.INFECTED)
```

The advantage of this design is that all simulation phases interact with ants only through these predicates. The foraging phase, for instance, is simply based on the list of ants satisfying `can_forage`. If we wanted to add a quarantine state in which an ant is kept away from foraging, we would only need to change `can_forage`, and all other phases would automatically respect the new rule.

5.2 The infection phase in full

The most complex part of the simulation is the daily infection phase, which implements all four sub-steps of Step 6 from the governing algorithm:

```
1 def _phase_infection(self):
2     if not self.cordyceps_enabled:
3         return
4
5     # On the introduction day, initialize the concentration.
6     if self.day == self.spore_intro_day:
```

```

7         self.spore_concentration = self.c0
8
9         # Nothing to do if there are no spores present.
10        if self.spore_concentration <= 0:
11            return
12
13        # (a) New infections: each healthy foraging adult is
14        # exposed to the environment and may pick up spores.
15        for ant in self._foraging_ants():
16            if ant.state == AntState.ADULT:
17                p_infect = min(self.spore_concentration
18                               * self.infection_rate_p, 1.0)
19                if self.rng.random() < p_infect:
20                    ant.transition_to(AntState.INFECTED)
21
22        # (b) Immune mini-game: no recovery. Each infected ant
23        # independently rolls to succumb or hold on.
24        for ant in self.ants:
25            if ant.state == AntState.INFECTED:
26                if self.rng.random() < self.p_succumb:
27                    ant.transition_to(AntState.ZOMBIE)
28
29        # (c) Fixed-duration timers advance each state.
30        for ant in self.ants:
31            if (ant.state == AntState.ZOMBIE
32                and ant.days_in_state >= Ant.ZOMBIE_DURATION):
33                ant.transition_to(AntState.SPORULATING)
34
35        for ant in self.ants:
36            if (ant.state == AntState.SPORULATING
37                and ant.days_in_state >= Ant.SPORULATION_DURATION):
38                ant.transition_to(AntState.DEAD)
39
40        # (d) Update spore concentration: decay plus new spores
41        # from currently sporulating ants.
42        n_sporulating = self._count_state(AntState.SPORULATING)
43        self.spore_concentration = (
44            self.spore_concentration * (1.0 - self.spore_decay_rate)
45            + self.spore_birth_rate * n_sporulating
46        )
47
48        # Set to zero if rounding makes it slightly negative.
49        self.spore_concentration = max(self.spore_concentration, 0.0)

```

6 Results and Discussion

The infection plots below use stacked filled regions to show how the total colony population is divided among healthy adults, infected ants, zombies, sporulating ants, larvae, and eggs over time. In these plots, the filled areas represent population categories, so the main information is the relative size of each category and how the total colony changes over time. When visual comparison alone

is not enough, we use ensemble statistics and RMSE values to make the conclusions quantitative.

6.1 Three qualitative infection regimes

Mild infection ($p = 0.08$): colony survives and recovers.

Figure 4 shows a lightly contaminated environment. The outbreak flares briefly near day 200: a few ants become infected, a few sporulating corpses appear, and spore concentration increases slightly. However, the outbreak then burns out. The feedback loop cannot sustain itself because, at $p = 0.08$, the rate of new infections is smaller than the rate at which sporulating ants expire and spores decay. Spores fall to near zero, and by day 300 the colony has returned to its normal seasonal rhythm. At the population level, the outbreak is barely visible.

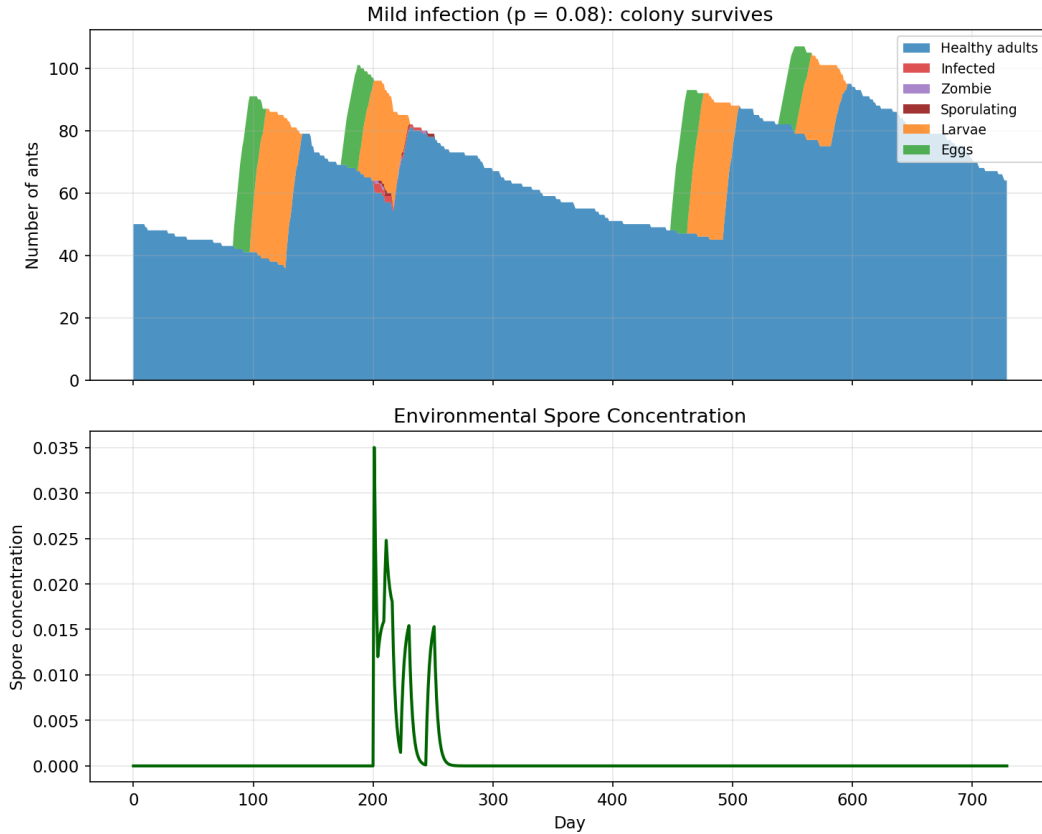


Figure 4: Mild infection ($p = 0.08$, $c_0 = 0.05$, spores introduced on day 200).

Moderate infection ($p = 0.18$): colony weakened, persists for a year, then collapses.

Figure 5 shows the intermediate case. After spore introduction, the infection neither fizzles nor explodes immediately. A persistent fraction of adults remain infected, and the spore concentration oscillates at a low but nonzero level. Eventually the spores die out, but not before the adult population has been reduced enough that foraging output can no longer sustain the larvae. The colony then collapses from starvation rather than from infection directly.

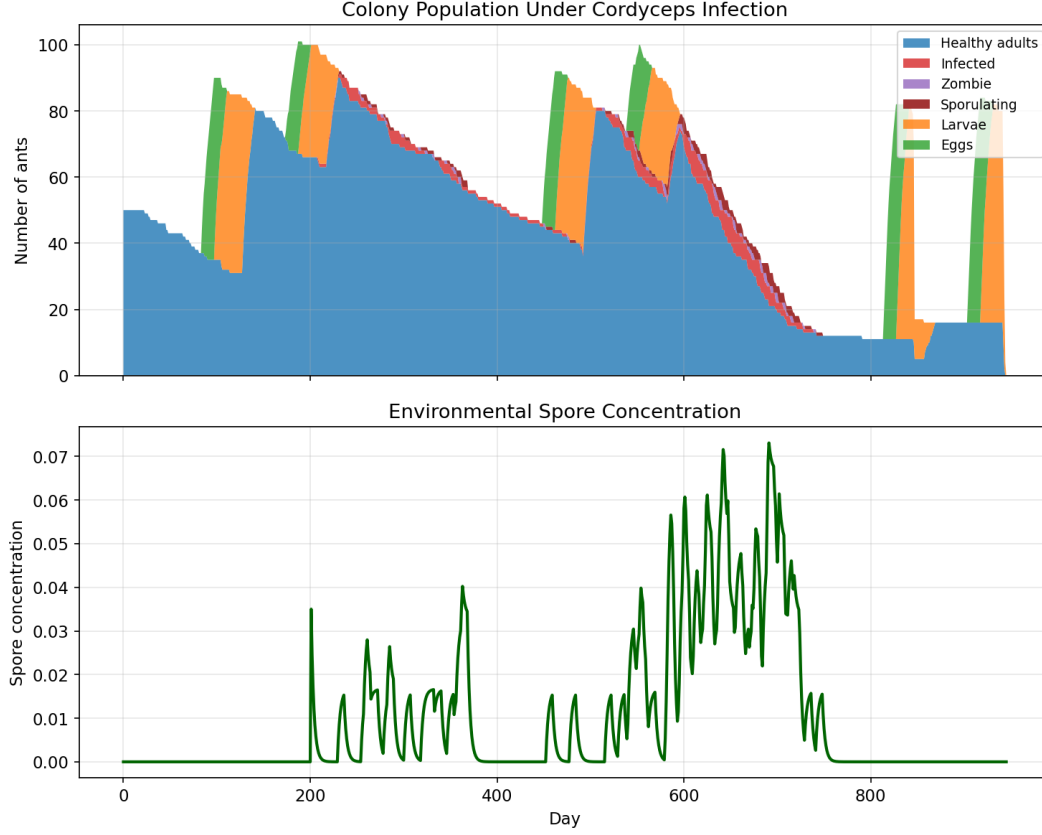


Figure 5: Moderate infection ($p = 0.18$, $c_0 = 0.05$, spores introduced on day 150).

Severe infection ($p = 0.35$): colony collapses.

Figure 6 shows the fatal case. After spore introduction on day 200, the feedback loop accelerates rapidly. Within a few months, the sporulating population is large enough to drive c_n to high values, which in turn infects nearly all remaining healthy foragers. Without foragers, food supply collapses. Eggs and larvae starve. The queen cannot lay new eggs. Total extinction occurs by around day 400–500. The spore concentration shows a sharp peak and then a crash because the colony itself was the fungus's resource, and once the colony is gone, the spores have nothing left to sustain them.

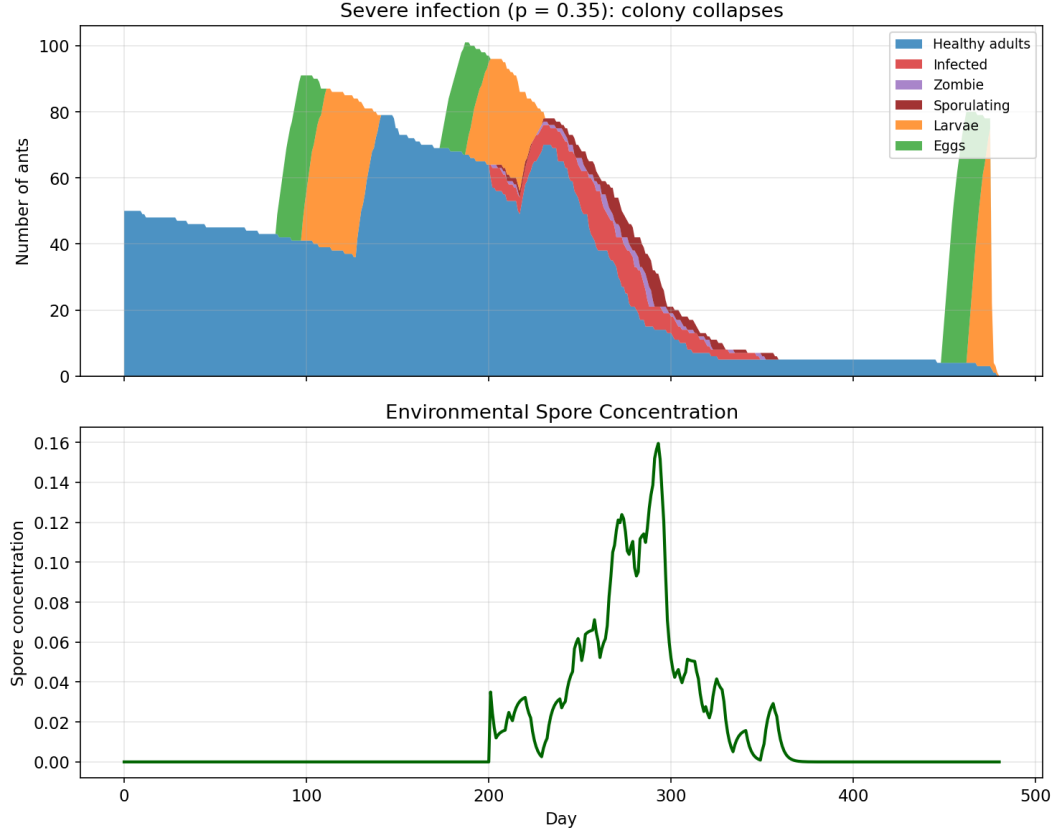


Figure 6: Severe infection ($p = 0.35$, $c_0 = 0.05$, spores introduced on day 200).

6.2 Parametric study: does initial colony size help?

The natural intuition is that a larger colony should be more resilient, since it can lose some members without losing essential functions. Figure 7 tests this intuition and shows the opposite: under the same infection rate ($p = 0.15$), a colony starting at $N_0 = 1000$ collapses faster and more reliably than one starting at $N_0 = 50$.

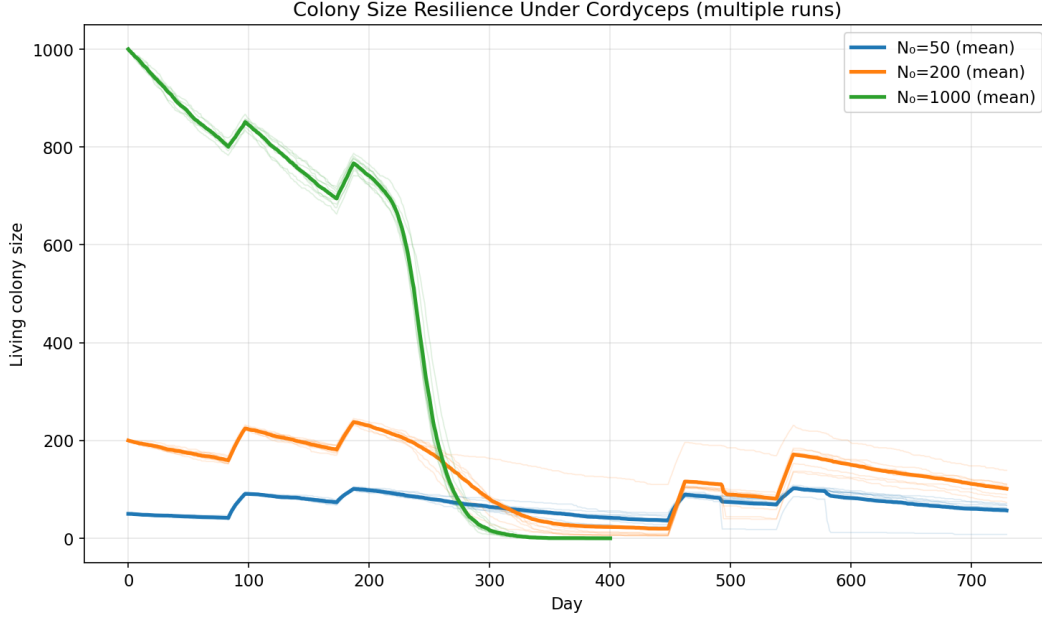


Figure 7: Living colony size versus time for three initial colony sizes, all at $p = 0.15$.

This is due to the $N^{-1/3}$ foraging model. A large colony sends out many foragers each day, exposing many ants to spores simultaneously. This produces a large initial infection wave that transitions into a large sporulating population, which drives c_n to high values before the colony can mount a demographic response. Meanwhile, each individual forager brings back less food at large N , so the food buffer that would support egg-laying during an outbreak is thinner. The small colony, by contrast, produces only a handful of sporulating ants at once. Spores never concentrate enough to sustain the loop, and the outbreak burns out.

6.3 Parametric study: how important is immune resistance?

Figure 8 sweeps the daily succumb probability p_s from 0.01 (mean 100 days to succumb; strong immunity) to 0.15 (mean about 6.67 days; weak immunity), measuring colony survival probability at day 730 averaged over 20 trials per value.

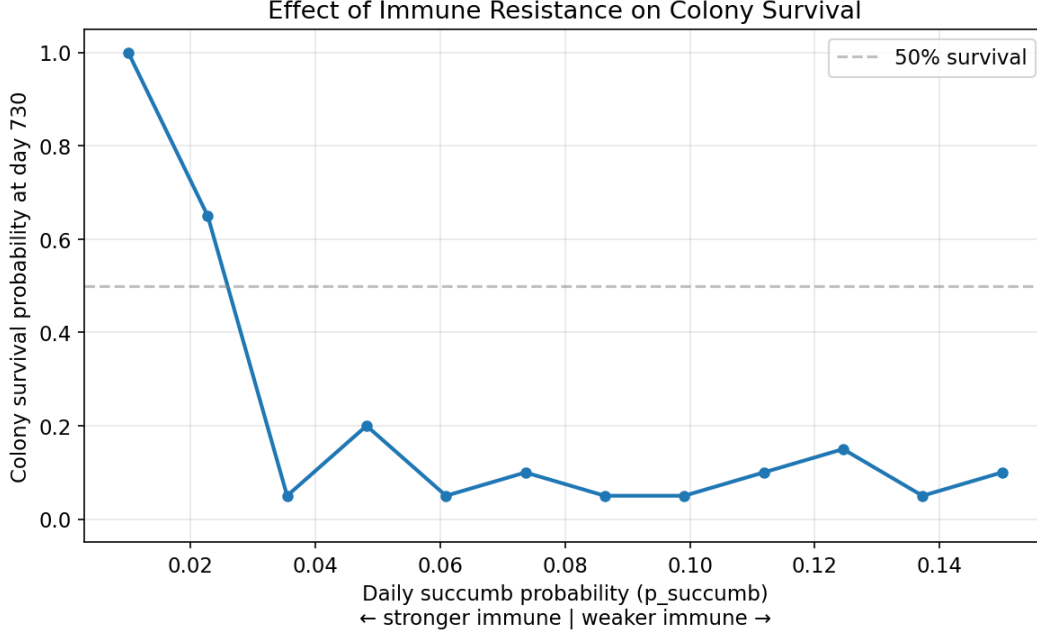


Figure 8: Colony survival probability at day 730 as a function of daily succumb probability p_s .

The transition from near-certain survival to near-certain extinction happens in a narrow band around $p_s \approx 0.02$ – 0.04 . At $p_s = 0.01$, the expected infected period is

$$\frac{1}{p_s} = \frac{1}{0.01} = 100 \text{ days.}$$

This delays the appearance of sporulating corpses, weakening the spore feedback loop. In many trials, the infection therefore fails to amplify before demographic noise, natural death, or food limitation reduces the number of infected ants. At $p_s = 0.05$, our baseline, the expected wait is only 20 days, which is short enough that many infected ants do succumb and sporulate. Above $p_s \approx 0.04$, survival probability is low but not exactly zero since stochastic variation occasionally lets the outbreak fizzle before it gains momentum.

6.4 Parametric study: threshold behavior over a range of infection rates

Infection prevalence, shown in Figure 9, is the time-averaged fraction of the living colony that is infected, measured over the post-introduction period. It stays near zero for small values of p and rises steeply above the threshold region near $p \approx 0.15$. In an endemic disease model, the critical infectivity marks the boundary below which the disease dies out in the growing population. Our simulation reproduces this threshold-like behavior without computing R_0 explicitly.

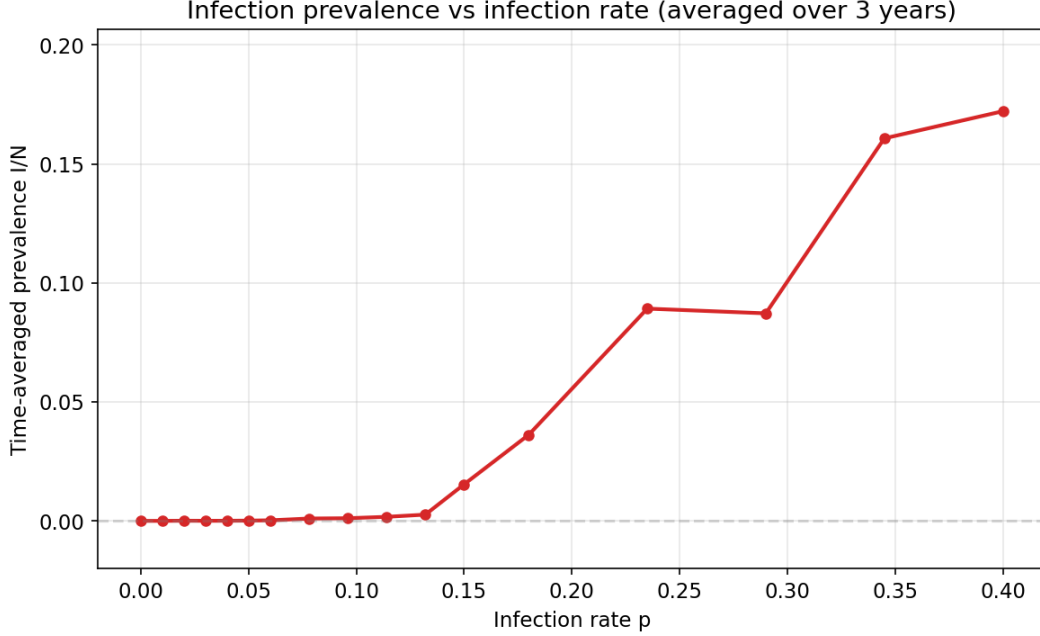


Figure 9: Time-averaged fraction of the colony that is infected versus infection rate p .

6.5 Stochastic vs. deterministic comparison

Figures 10 and 11 compare 30 stochastic trials against the deterministic ODE system (1)–(8) at two colony sizes. We make this comparison precise by comparing the deterministic solution to the stochastic ensemble mean, not to individual stochastic trials.

For any model variable Y , for example living colony size, infected count, food, or spore concentration, let $\bar{Y}_{\text{stoch}}(n)$ be the average across stochastic trials on day n , and let $Y_{\text{det}}(n)$ be the ODE value on the same day. We compute the root-mean-square error

$$\text{RMSE}_Y = \sqrt{\frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} (\bar{Y}_{\text{stoch}}(n) - Y_{\text{det}}(n))^2}.$$

This metric is useful because it separates two questions: whether the deterministic curve follows the average stochastic behavior, and whether the stochastic trials have large variance around that average.

The RMSE values are shown in Table 4. To make the calculation reproducible, the stochastic runs used for this table were generated using the same parameter values stated in the figure captions, and the deterministic curve was generated directly from equations (1)–(8). The RMSE calculation was then applied to the saved daily arrays, not estimated visually from the plots.

These values were computed using 30 stochastic trials with random seeds $0, 1, \dots, 29$. The deterministic curve was computed by forward Euler with the same one-day timestep as the stochastic simulation. For the $N_0 = 50$ case, the initial food was $F_0 = 200$. For the $N_0 = 500$ case, the

initial food was $F_0 = 2000 = 4N_0$, matching the scaled-food convention used in the colony-size comparison.

Case	Window	RMSE living	RMSE infected	RMSE spores	RMSE food	Final extinction probability
$N_0 = 50$	days 200–730	10.44	2.21	0.0123	74.71	0.00
$N_0 = 500$	days 200–730	63.31	5.77	0.0478	360.89	1.00
$N_0 = 500$	days 200–276	20.05	11.56	0.1012	244.38	1.00

Table 4: Quantitative comparison between the deterministic ODE solution and the stochastic ensemble mean. The pre-crash window for $N_0 = 500$ ends at day 276, just before the stochastic ensemble mean falls below 20% of its initial population. The food RMSE is relatively large because the stochastic simulation applies the food cap before eating and includes starvation, while the ODE uses a smooth expected food balance.

The RMSE values should be interpreted as consistency checks rather than as proof that the deterministic and stochastic models are identical. The ODE is a smooth average model that can contain fractional numbers of ants, while the stochastic model counts individual ants, uses fixed state timers, includes random foraging and starvation, and treats extinction as permanent. Therefore, RMSE is most useful before the colony becomes very small. After extinction or near-extinction, disagreement is expected because the stochastic model cannot recover once all living ants are gone, while the ODE can still have small nonzero values such as 0.2 ants or 0.5 eggs.

The infected and spore RMSE values are also sensitive to timing differences: the stochastic model uses fixed individual delays, while the ODE uses continuous transition rates. A small shift in the timing of the infection peak can create a noticeable RMSE even when the two models show the same qualitative outbreak pattern.

At $N_0 = 50$ (Figure 10), the models do not align closely, and this discrepancy is expected. A colony of 50 ants is small enough that individual random events matter. Losing only a few foragers can change the food balance, and one or two early sporulating ants can decide whether the feedback loop survives. The deterministic model also permits fractional populations, while the stochastic model has a hard zero: once the last ant dies, the colony cannot recover. Therefore, for small colonies, the deterministic model is best interpreted as a smooth reference curve, not as a prediction of any individual stochastic trajectory.

At $N_0 = 500$ (Figure 11), the deterministic solution is closer to the stochastic ensemble mean in relative terms, especially before the colony approaches extinction. The full-window living-population RMSE is 63.31 ants, which is about 12.7% of the initial population. Over the pre-crash window, the living-population RMSE drops to 20.05 ants, or about 4.0% of the initial population. This is expected because random individual events average out more when the colony is large. As a result, the average of the stochastic simulations is closer to the deterministic curve before the colony becomes very small.

However, the infected and spore variables still show noticeable differences even before the crash. This is not surprising because the stochastic model uses fixed individual timers for eggs, larvae, zombies, and sporulating ants, while the ODE replaces these timers with continuous exponential-style transition rates. In other words, the two models are not exact copies of one another. They agree best for broad population trends, not for the exact timing of infection peaks.

The remaining discrepancy after the crash comes from a basic difference between the two models. Once a stochastic colony reaches zero living ants, extinction is permanent. The ODE, however, can still contain fractional ants, fractional eggs, and fractional infected individuals. Those fractions can persist or regrow if the smooth food and reproduction terms allow it. Therefore, some mismatch in Figures 10 and 11 is expected. The stochastic model counts whole ants and cannot recover once all living ants are gone, while the ODE model uses smooth average values that can remain slightly above zero. The comparison is still useful because the deterministic curve shows the average trend before extinction, while the stochastic model shows the permanent collapse that occurs when the colony reaches zero living ants.

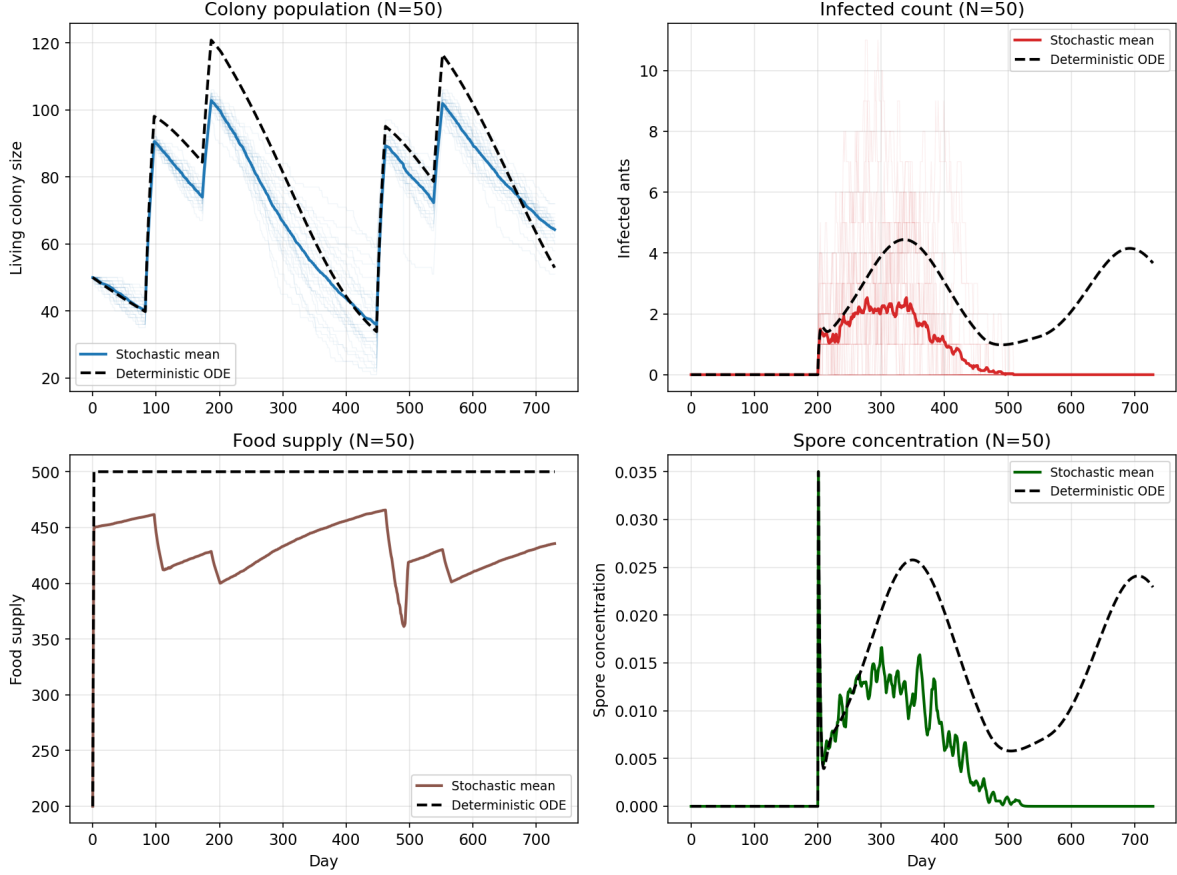


Figure 10: Stochastic vs. deterministic comparison at $N_0 = 50$, $p = 0.15$. The mismatch is expected because stochastic extinction, integer ant counts, and random food draws dominate small-population behavior.

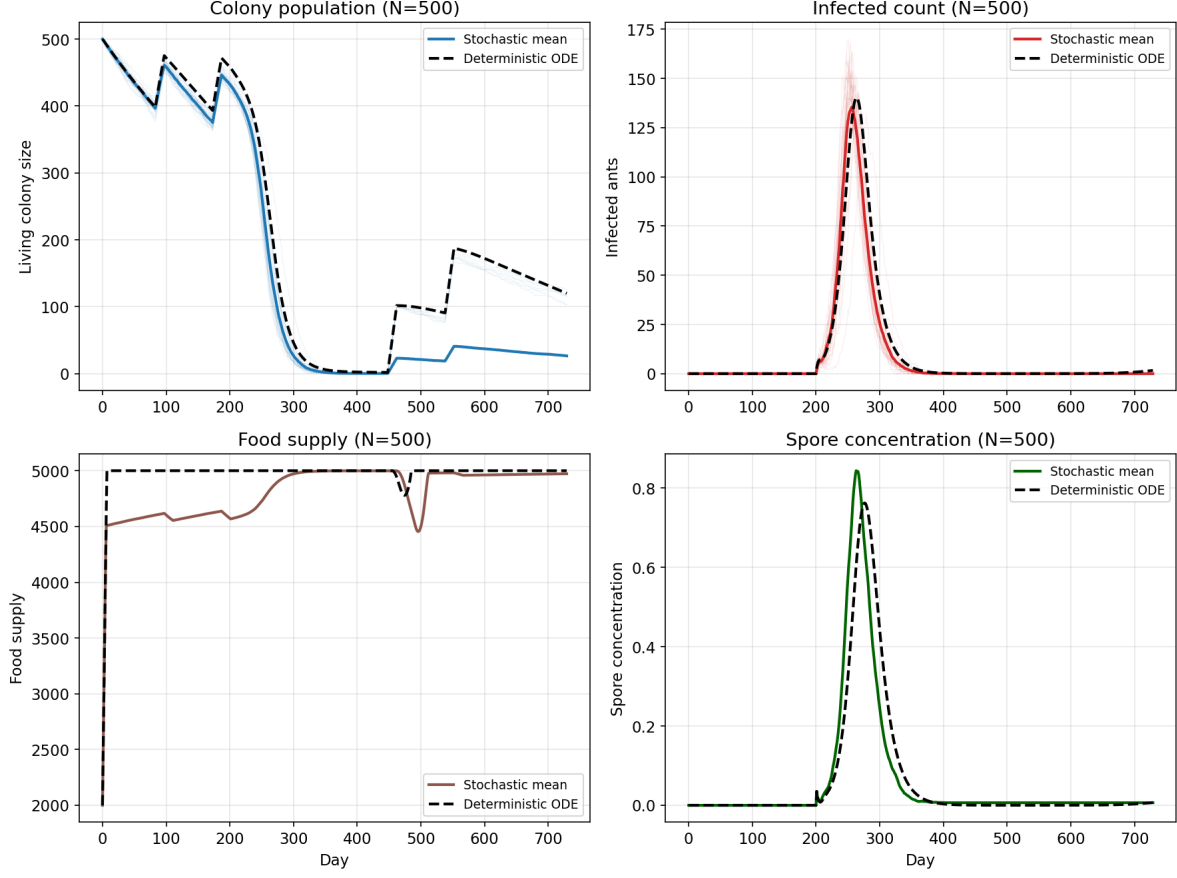


Figure 11: Stochastic vs. deterministic comparison at $N_0 = 500$, $p = 0.15$. The stochastic mean and ODE are expected to agree best before the extinction threshold; after the crash, fractional ODE populations and discrete stochastic extinction lead to divergence.

7 Conclusions

This project developed an agent-based stochastic simulation of *Ophiocordyceps unilateralis* infecting a carpenter ant colony to determine what controls colony survival during an outbreak. For the parameter ranges tested, the model shows threshold-like behavior near $p \approx 0.15$, marking a transition between recovery and collapse. Below this value, sporulation and new infections cannot sustain one another, so the outbreak dies out. Above it, the feedback loop becomes self-sustaining and drives the colony toward extinction.

The model also shows that larger colonies are not necessarily more resilient under the $N^{-1/3}$ foraging model. Larger colonies send out more foragers, so they can generate larger infection waves when spores are present. These infection waves can overwhelm the system before reproduction and food storage can compensate. The simulations where p_s is varied also show that colony survival is highly sensitive to the daily probability that an infected ant becomes a zombie. When p_s increases from about 0.01 to 0.04, colony survival shifts from near certainty to near zero, suggesting that the timing of infected-to-sporulating progression is one of the most important controls in the simulation.

Finally, the deterministic ODE is useful as a smooth average comparison for the stochastic model, especially when the colony is still large. However, it is not expected to match small colonies or post-crash behavior because it allows fractional ants and cannot represent permanent stochastic extinction. The RMSE values make this comparison more precise: for the large-colony case, the living-population RMSE is much smaller before the crash than over the full time window. This shows that most of the disagreement happens after the stochastic colony becomes very small or goes extinct, which is an expected modeling difference.

References

- [1] Evans, H. C., Elliot, S. L., and Hughes, D. P. “Ophiocordyceps unilateralis: A keystone species for unraveling ecosystem functioning and biodiversity of fungi in tropical forests?” *Communicative & Integrative Biology*, 4(5):598–602, September 2011. doi: 10.4161/cib.16721.
- [2] Peskin, Charles S. *Introduction to Computer Simulation*. Lecture notes, Courant Institute of Mathematical Sciences, New York University. Copyright 2020, 2021, 2023, 2025.
- [3] Hughes, D. P., Andersen, S. B., Hywel-Jones, N. L., Himaman, W., Billen, J., and Boomsma, J. J. “Behavioral mechanisms and morphological symptoms of zombie ants dying from fungal infection.” *BMC Ecology*, 11:13, 2011. doi: 10.1186/1472-6785-11-13.
- [4] Crawley, S., and Hayes, C. C. “Biology and Control of Carpenter Ants.” NC State Extension Publications, reviewed/revised February 7, 2023.
- [5] Orkin. “Carpenter Ant Life Cycle.” Orkin Pest Library, accessed 2026.